Supplemental Document

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Polarization imaging-based radiomics approach for the staging of liver fibrosis: supplement

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Mueller matrix contains full polarization information of pathological samples, but the individual elements often lack explicit connections to specific microstructures. Therefore, various polarization parameters [1, 2] with more explicit physical meanings were derived from the Mueller matrix to characterize the microstructures of pathological samples. In 1996, Lu and Chipman [3] presented the Mueller matrix polar decomposition (MMPD) method to characterize the polarization properties of a medium by decomposing a Mueller matrix into three matrix factors including a diattenuator, a retarder, and a depolarizer. This study used the MMPD diattenuation parameter D. In 2008, Ghosh *et al*. [4] investigated the efficacy of an MMPD method in the characterization of biological samples. The involved parameters including linear retardance parameter δ, linear retardance orientation angle parameter θ, optical rotation parameter Ψ, and the depolarization coefficient Δ.

To characterize the polarization properties of a medium, besides Mueller matrix polar decomposition, Mueller matrix transformation (MMT) is a general concept [1, 5] that aims at transforming the Mueller matrix into new polarization parameters to describe certain physical properties of the medium. In 2013, He *et al*. [1, 5] presented the MMT method to derive new quantitative polarization parameters through the trigonometric functions fitted Mueller matrix elements in the characterization of the microstructural features of complex biological samples. The MMT parameters used in this study include anisotropy degree parameter t_1, polarizance parameter b, circular birefringence parameter β, normalized anisotropy parameter A and local orientation parameter α. In 2018, Li *et al*. [6] proposed a set of polarization parameters that are invariant to the azimuth orientation of the sample. The Mueller matrix rotation invariant (MMRI) parameters are linear polarizance parameter P_L, linear diattenuation parameter D_L, and the birefringence related parameters q_L and r_L. The corresponding orientation angle parameters are linear polarizance orientation angle parameter α_P, linear diattenuation orientation angle parameter α_D, and the birefringence related orientation angle parameters α_q and α_r.

In 2021, Dong *et al*. [2] compared an asymmetric anisotropic Mueller matrix with a pure linear retarder Mueller matrix and a pure linear diattenuator Mueller matrix to derive Mueller matrix linear birefringence identity (MMLBI) parameters and Mueller matrix linear diattenuation identity (MMLDI) parameters. In a pure linear retardance system and a pure linear diattenuation system, the sum of some Mueller matrix elements equals a constant. For instance, the sum of m_{34} and m_{43} equals 0. Therefore, MMLBI and MMLDI parameters are all equal to 0 in a pure linear retardance system or a pure linear diattenuation system respectively. The breaking of transpose symmetry of a Mueller matrix often occurs in pathological tissues. The absolute values of the MMLBI and MMLDI parameters describe the deviation of a tissue sample from a pure single anisotropic system.

Dong *et al*. [7] organized previously derived polarization parameters from different physical methods into Polarimetry Basis Parameters (PBPs). This study also employed the term Polarimetry Basis Parameter (PBP) to characterize the microstructures of liver fibrosis tissues. Table S1 of the supplemental document summarized the calculation equations and the physical meanings of the 26 PBPs used in this study. The 26 PBPs including Mueller matrix polar decomposition (MMPD) parameters [1, 3, 4] δ, θ, D, Δ and Ψ, Mueller matrix transformation (MMT) parameters [1, 5] t_1, b, β, A and α, Mueller matrix rotation invariant (MMRI) parameters [1, 6] P_L, D_L, q_L and r_L, Mueller matrix azimuth orientation (MMAO) angle parameters [1, 6] α_P, α_D, α_q and α_r, Mueller matrix linear birefringence identity (MMLBI) parameters [1, 2] P_1, P_2, P_3 and P_4, and Mueller matrix linear diattenuation identity (MMLDI) parameters [1, 2] P_5, P_6, P_7 and P_8.

Table S1. Polarimetry Basis Parameters (PBPs)

PBPs			
MMPD [1, 3, 4]	Physical Meaning	MMT [1, 5]	Physical Meaning
$\delta = \cos^{-1}\left\{\left[\begin{pmatrix} M_R(2,2)+ \\ M_R(3,3) \end{pmatrix}^2 + \begin{pmatrix} M_R(3,2)+ \\ M_R(2,3) \end{pmatrix}^2\right]^{\frac{1}{2}} - 1\right\}$	Linear retardation	$t_1 = \frac{\sqrt{(m22-m33)^2+(m23+m32)^2}}{2}$	Anisotropy degree
$\theta = 0.5\tan^{-1}\left(\frac{r_2}{r_1}\right)$	Linear retardation orientation angle	$b = \frac{m22+m33}{2}$	Polarizance
$D = \sqrt{(m12)^2+(m13)^2+(m14)^2}$	Diattenuation	$\beta = \frac{\lvert m23-m32\rvert}{2}$	Circular birefringence
$\Delta = 1 - \frac{\lvert tr(M_\Delta - 1)\rvert}{3}$	Depolarization	$A = \frac{2b\cdot t_1}{b^2+t_1^2}$	Normalized anisotropy
$\psi = \frac{1}{2}\tan^{-1}\left[\frac{M_R(3,2)-M_R(2,3)}{M_R(2,2)-M_R(3,3)}\right]$	Optical rotation	$\alpha = \frac{1}{4}\arctan 2\begin{pmatrix} m23+m32, \\ m22-m33 \end{pmatrix}$	Local orientation
MMRI [1, 6]	Physical Meaning	MMLBI [1, 2]	Physical Meaning
$P_L = \sqrt{(m21)^2+(m31)^2}$	Linear polarizance	$P_1 = \frac{m43+m34}{2}$	Deviation of an asymmetric anisotropic Mueller matrix compared with a pure linear retarder Mueller matrix
$D_L = \sqrt{(m12)^2+(m13)^2}$	Linear diattenuation	$P_2 = \frac{m42+m24}{2}$	
$q_L = \sqrt{(m42)^2+(m43)^2}$	Capability of transforming between linear and circular polarizations	$P_3 = \lvert q_L - r_L\rvert$	
$r_L = \sqrt{(m24)^2+(m34)^2}$		$P_4 = \frac{m34\cdot m24 - m42\cdot m43}{2}$	
MMAO [1, 6]	Physical Meaning	MMLDI [1, 2]	Physical Meaning
$\alpha_P = \frac{1}{2}\arctan 2(m31, m21)$	Orientation of maximum transmittance	$P_5 = \frac{m12-m21}{2}$	Deviation of an asymmetric anisotropic Mueller matrix compared with a pure linear diattenuator Mueller matrix
$\alpha_D = \frac{1}{2}\arctan 2(m13, m12)$		$P_6 = \frac{m13-m31}{2}$	
$\alpha_q = \frac{1}{2}\arctan 2(m42, -m43)$	Orientation of fast axis	$P_7 = \lvert D_L - P_L\rvert$	
$\alpha_r = \frac{1}{2}\arctan 2(-m24, m34)$		$P_8 = \frac{m12\cdot m13 - m21\cdot m31}{2}$	

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